

Sample Question

1. Explain the difference between linear data structures and non-linear data structures. [2]
2. Draw a simple diagram of a singly linked list with 5 nodes and indicate the head and tail. [3]
3. Consider the array, **A** = [12, 45, 23, 67, 34]. Perform the following: [2+3]
 - a) Find the position of 67 using linear search. Show each comparison.
 - b) Before performing binary search, sort the array in ascending order. Then find the position of 45 using binary search. Show all steps.
4. Consider the array, **A** = [29, 10, 14, 37, 13]. [3+2]
 - a) Sort the array using Selection sort. Show all passes.
 - b) Write Pseudocode or algorithm for bubble sort.
5. Consider a stack implemented using an array with maximum size 5. Initially, the stack is empty. [3+2]
 - a) Perform the following operations and show the stack after each step:
Push(10), Push(20), Push(30), Pop(), Push(40), Pop(), Push(50)
 - b) Explain how to handle stack overflow and underflow. Give examples for each case.